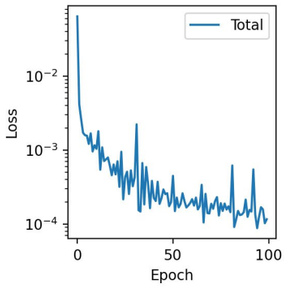
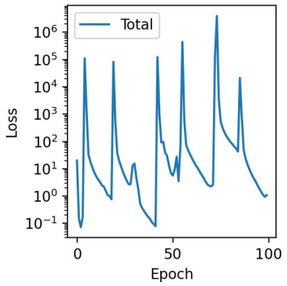
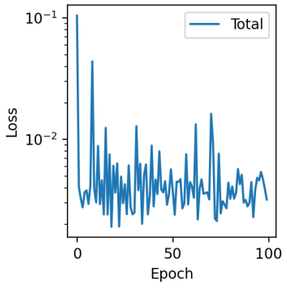
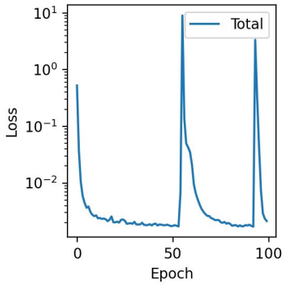
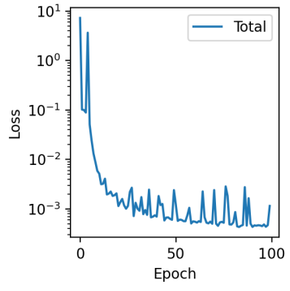
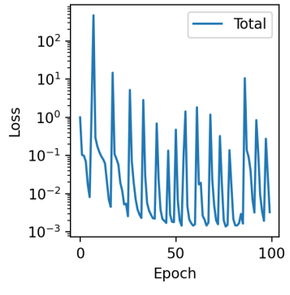
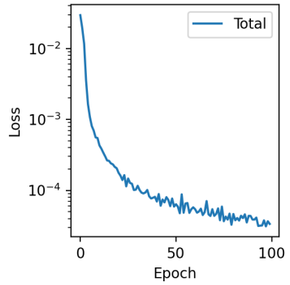
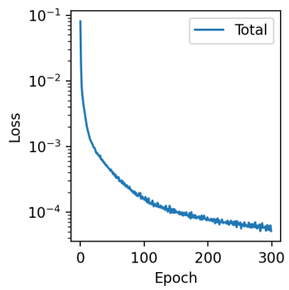
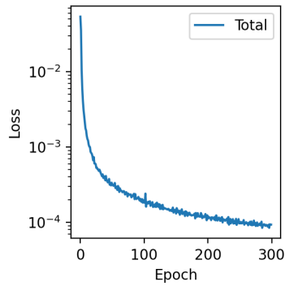
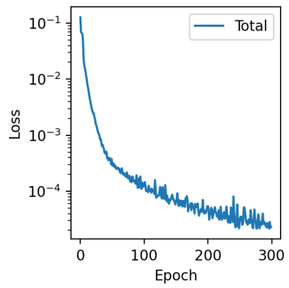
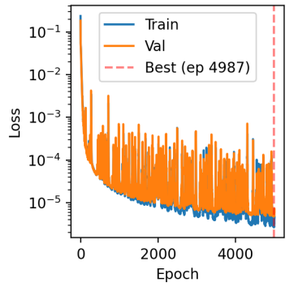
Summary of Convolutional Autoencoder Runs (3D Case, CDI Method, Diffuse Interface)

Case	Input size and channels $N_c \times N_x \times N_y \times N_z$	Sample size N_{TR} N_{VL}	Architecture Base dim `dim_mult` Channel output	Latent space CR	Learning rate	Epochs	ETR	Notes / Result
1	$2 \times 64 \times 64 \times 64$ G, V	390 312|78	16 (1) (16, 20)	$20 \times 64 \times 64 \times 64$ $N_z = 20$ CR=0.1	1×10^{-3}	100	~ 4 h	
2	$2 \times 64 \times 64 \times 64$ G, V	390 312|78	16 (1,2,4,8) (16,32,64,20)	$20 \times 16 \times 16 \times 16$ $N_z = 20$ CR=6.4	1×10^{-3}	100	~ 9 h	
3	$1 \times 64 \times 64 \times 64$ G	390 312|78	16 (1,2,4,8) (16,32,64,20)	$20 \times 16 \times 16 \times 16$ $N_z = 20$ CR=3.2	1×10^{-3}	100	~ 2 h	
4	$1 \times 64 \times 64 \times 64$ G	390 312|78	16 (1,2,4,8,16,32,64,128) (16,32,64,128,256,512,1024,10)	$10 \times 1 \times 1 \times 1$ $N_z = 10$ CR=26214.4	1×10^{-4}	100	~ 2 h 2×V100 32GB GPU	

Case	channels and Input size $N_c \times N_x \times N_y \times N_z$	Sample size N_{TR} N_{VL}	Architecture	Latent space CR	Learning rate	Epochs	Notes	Result
5 3DMPF datasets	$1 \times 128 \times 256 \times 128$ G	918 734|184	16 (1,2,4,8,16,32,64) (16,32,64,128,512,16)	$16 \times 4 \times 8 \times 4$ $N_z = 16$ CR=2048	1×10^{-4}	100	12hrs A100 80GB Batch size = 2	
6 3DMPF datasets	$1 \times 128 \times 256 \times 128$ G	918 734|184	16 (1,2,4,8,16,32,64) (16,32,64,128,512,16)	$16 \times 4 \times 8 \times 4$ $N_z = 16$ CR=2048	1×10^{-4}	100	23hrs A100 80GB Batch size = 4	
7 3DMPF datasets	$1 \times 128 \times 256 \times 128$ G	918 734|184	16 (1,2,4,8,16,32,64) (16,32,64,128,512,16)	$16 \times 4 \times 8 \times 4$ $N_z = 16$ CR=2048	1×10^{-5}	100	23hrs A100 80GB Batch size = 4	
8 2DMPF datasets	$1 \times 1 \times 512 \times 256$ G	2400 1920|480	16 (1,2,4,8,16,32,64,128) (16,32,64,128,256,512,1024,4)	$4 \times 8 \times 4$ $N_z = 4$ CR=1024	1×10^{-5}	100	5hrs A100 40GB Batch size = 16	

Case	channels and Input size $N_c \times N_x \times N_y \times N_z$	Sample size N_{TR} N_{VL}	Architecture	Latent space CR	Learning rate	Epochs	Notes	Result
9 2DMPF datasets	$3 \times 512 \times 256 \times 1$ G, V, W	2400 1920 480	Resent-CAE Base dim=16 8 Layers (1,2,4,8,16,32,64,128) (16,32,64,128,256,512,1024,4)	$4 \times 8 \times 4$ CR=3072	1×10^{-5}	100	5hrs A100 80Gb Batch size = 16	
10 2DAdvection datasets	$1 \times 256 \times 256 \times 1$ u	$79 \times 16 = 1264$ $15 \times 16 = 240$ $64 \times 16 = 1024$	Resent-CAE Base dim=16 6 Layers (1,2,4,8,16,32) (16,32,64,128,256,4)	$4 \times 16 \times 16$ CR=64	1×10^{-5}	300	1.5hrs T100 8Gb Batch size = 8	
11 2DAdvection datasets	$1 \times 256 \times 256 \times 1$ u	$79 \times 16 = 1264$ $15 \times 16 = 240$ $64 \times 16 = 1024$	Wang (simple)-CAE Base dim=16 6 Layers (1,2,4,8,16,32) (16,32,64,128,256,512)	4 CR=16384	1×10^{-4}	5000	3hrs T100 8Gb Batch size = 8 early stop, 500	

0.1 MPF Sample Dataset: 10 Simulations (Phase-Field Model)

In total, there are **500 simulations**, each containing **51 snapshots**.

Case properties:

$$X = [\rho^*,\, \mu^*,\, Bo,\, Re], \qquad Y = [c,\, u,\, v,\, w,\, P,\, \rho]$$

Property ranges:

$$\rho^* \in [10,\, 1000], \qquad \mu^* \in [1,\, 100], \qquad Bo \in [10,\, 500], \qquad Re \in [10,\, 500]$$

3D dataset shape:

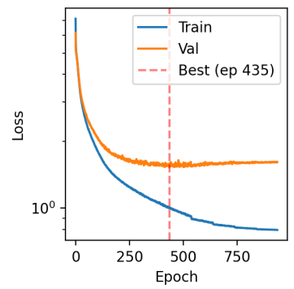
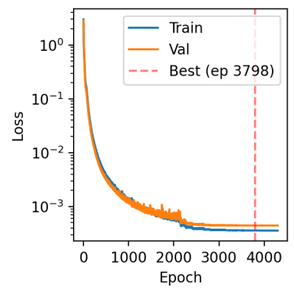
$$[n_t = 10,\ \text{channels} = 6,\ n_x = 256,\ n_y = 256,\ n_z = 256]$$

The actual spatial resolution

$$[128,\ 256,\ 128]$$

Key questions:

1. Should we consider the entire domain, or only the box surrounding the bubble? The entire domain
2. If we use the full domain, should the snapshots be fed to the autoencoder in batches or as a single input block? as a single block
3. Which bubbles should be included during training? Should cases involving bubble breakup also be considered? DR and VR are fixed (=10, as in 3D cases we have) and vary the Re and BO numbers
4. After training a well-performing autoencoder, should we test it on our own dataset?

Case	Physical parameters	Sample size N_{TR} N_{VL}	Architecture	Latent space from CAE	Learning rate	Epochs	ETR	Notes / Result
9* 2DMPF datasets	5 (DR, VR, Re, Eo, t)	2400 1920 480	H=64 3 Layers	$4 \times 8 \times 4$ CR=64	1×10^{-3}	5000	20 secs T100 8Gb Batch size = 32 early stop, 500	
10* 2DAdvection datasets	2 (μ, t)	$79 \times 16 = 1264$ $15 \times 16 = 240$ $64 \times 16 = 1024$	H=64 3 Layers	$4 \times 16 \times 16$ CR=64	1×10^{-3}	5000	20 secs T100 8Gb Batch size = 32 early stop, 500	

Summary of CNN (MLP) Runs